Midterm Part II* (Time Allowed: 1 Hour)**

Submit one cpp file with comments of start of second question.

1. Create a class named Course' with data members: string course code, string course title, integer credit hours, and string pre requisite course code. Also add static member number of courses. Write following member function:

- non-parameterized constructor. Assign 3 to credit hours and course code according to number of courses. Assign "Introduction to Computers" to title and assign empty string to remaining members
- parameterized constructor with single parameter credit hours. Check if credit hours are less than 1 assign 3 to credit hours, assign code according to number of course, assign title "Introduction to Computers" and assign empty string to remaining members
- set function with single parameter ingle parameter credit hours. Check if credit hours are less than 1 assign 3 to credit hours, assign code according to number of course, assign title "Introduction to Computers" and assign empty string to remaining members
- overload stream insertion operator (see example output below)
- write getter to get number of courses
- write appropriate destructor
- do appropriate handling of static member no of courses

In main create object of class Course using single parameterized constructor with credit hours 2. Print object using cout. Next ask user to enter a number. Create dynamic array of courses according to number. Call setter for array (to set credit hours) and print values using cout. Print number of courses. Delete dynamic array of students. Create a dynamic single object. Call setter for your dynamic object. Again print number of courses. Finally delete dynamic object.

Note: Avoid duplication of code to get maximum numbers.

Example Output:
Course Code: 3
Title: Probability and Statistics
Credit Hours: 3
Pre Requisite: NIL (if there is no pre requisite course)

2. Overload following functions and operators in class Mountains: (You may change variable names)

```
class Mountains{
 int noOfMountains;
 float *maxHeight;
 float *minheight;
  ...
};
```

- Copy constructor
- Assignment operator
- += (float) to add height in both max & min height and return current object
- == Return true, if first object and second object has equal average max height, otherwise return false
- \+ Create new object. merge first & second object and return new object
- Destructor
- Overload stream insertion operator to print Mountains object in following format: (do required formatting. Both max & min heights are in range 10 to 9999)

```
No of Mountains: 3
Max Height:  50    200   1500
Min Height : 15     90    800
```